Aliah University

Spring Semester Examination - 2025

BCA (CSE) 2nd year, 4th Semester Examination

Paper Name: Probability and Statistics **Full Marks: 80**

Paper Code: MATUGBS05 **Time: 3 hr**

1.	**Group-A (Answer all questions)**	**2 x 5 = 10**	
a.	Discuss the discrete and continuous Random experiments with suitable examples.	2	(CO2, Und)
b.	What's the probability that you roll off a die and will get 4 or higher?	2	(CO1, App)
c.	What is Gamma Distribution, mathematically?	2	(CO3, Und)
d.	Justify the joint distribution in probability theory.	2	(CO4, Ana)
e.	The number of ships to arrive at a harbor on any given day is a random variable represented by x. The probability distribution for x is: x: 21, 22, 23, 24, 25; P(x) 0.6, 0.7, 0.3, 0.5, 0.1; find probablity of $x \le 24$	2	(CO1, Eval)
	Group-B (Answer any Five (05) questions)	**5 x 6 = 30**	
2.	Explain all the Probability Properties. Define Probability Mass Function and Probability Density Function, mathematically.	2+2 +2	(CO2, Und), (CO2, Und)
3.	Explain (β_1, γ_1) and (β_2, γ_2) in statistics and discuss their characteristics.	3+3	(CO3, Ana)
4.	The following table gives production yield in kg. per hectare of wheat of 150 farms in a village. Production yield (kg. per hectare): 50–53, 53–56, 56–59, 59–62, 62–65, 65–68, 68–71, 71–74, 74–77 . Number of farms: 3, 8, 14, 30, 36, 28, 16, 10, 5 . Calculate the mean, median and mode values.	2+2 +2	(CO4, App & Ana)
5.	Find the mean and variance for the Poisson Distribution. What is the Central Limit Theorem?	2+2 +2	(CO3, App & Und)
6.	The first four moments of a distribution are 1,4, 10 and 46 respectively. Compute the moment coefficients of skewness and kurtosis and comment upon the nature of the distribution. Give formula for Spearman's Rank Correlation (ρ).	4+2	(CO3, Ana, Eval)
7.	Compute the Karl Pearson's coefficient of skewness from the following data : Daily Expenditure (Rs.): 0-20, 20-40, 40-60, 60-80, 80-100 & No. of families: 13, 25, 27, 19, 16. What are Coefficients of Alienation and of Non-Determination?	4+2	(CO4, App & Ana)

Group-C (Answer any Four (04) questions)		4 x 10 = 40	
8.	Derive the Mean and Variance for the discrete Random experiment. A multiple-choice exam has 12 questions, each with 4 options, only one of which is correct. A student guesses on all questions. Find the mean and variance of the number of correct answers using Binomial distributions. Find the rank correlation coefficient for the percentage of marks secured by a group of 8 students in Economics and Statistics. X: 50, 60, 65, 70, 75, 40, 70, 80 & Y: 80, 71, 60, 75, 90, 82, 70, 50	4+2 +4	(CO1, App & Ana)
9.	The following measures were computed for a frequency distribution: Mean = 50, coefficient of Variation = 35% and Karl Pearson's Coefficient of Skewness = - 0.25. Compute Standard Deviation, Mode and Median of the distribution. In a batch of 15 bulbs, each has a 0.1 probability of being defective. Find the probability that at most 2 bulbs are defective. Let the joint probability density function (pdf) of continuous random variables X and Y be given by: $f(x, y) = 6xy$, for $0 \leq x \leq 1$ and $0 \leq y \leq 1$; 0, otherwise. (a) Verify that $f(x, y)$ is a valid probability density function (pdf). (b) Find the marginal pdfs of X and Y. (c) Find $P(X \leq 0.5, Y \leq 0.5)$.	4+6	(CO2, App) (CO4, Eval)
10.	Compute the first four central moments with the following data. Also find the two beta coefficients. Value: 5, 10, 15, 20, 25, 30, 35 & Frequency: 8, 15, 20, 32, 23, 17, 5. Discuss the concept for coefficient of concurrent deviations. Calculate coefficient of correlation by the method of concurrent deviation from the following data. X: 60, 59, 72, 51, 55, 54, 65 & Y: 23, 36, 10, 38, 33, 44, 33.	6+1 +3	(CO3, App) (CO4, Ana)
11.	Find the mean and variance for the Normal Distribution. Calculate Karl Pearson's coefficient of correlation from the following data using 44 and 26 respectively as the origin of x and y and also discuss the nature of correlation coefficient. x: 43, 44, 46, 40, 44, 42, 45, 42, 38 40 42 57 . y: 29, 31, 19, 18, 19, 27, 27, 29, 41, 30, 26, 10	4+6	(CO3, App) (CO3, Eval)
12.	Calculate Bowley's coefficient of Skewness. Weekly Wages (Rs.): Below 200, 200-250, 250-300, 300-350, 350-400 400 & above. No of Workers(f): 10, 25, 145, 220, 70, 30. The joint probability distribution of two discrete random variables X and Y is given as follows: The probability that X = 1 and Y = 1 is 0.3; The probability that X = 1 and Y = 2 is 0.2; The probability that X = 2 and Y = 1 is 0.1; The probability that X = 2 and Y = 2 is 0.4. Using this joint distribution, compute the following: a) The expected values E(X) and E(Y). b) The variance of X and the variance of Y. c) The expected value of the product E(XY).	5+5	(CO3, Eval) (CO4, Eval)